Tabnine: Using Google Cloud powered AI to help developers code faster

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Tabnine uses Go
to power its AI sc
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workflows.

About Tabnine

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Tabnine's goal is to create and deliver a top-to-bottom AI-assisted development workflow that empowers all code creators, in all languages, from concept through to completion.

Industries: Technology

Location: Israel

Cloud IAM (https://cloud.google.com/iam/), Vertex AI (https://cloud.google.com/vertex-ai/)

Google Cloud results

- Makes coding best

Helps r
1 millic

practices available in code time
- Speeds up on-boarding process for new developers
- Improves coding quality for more than 1 million users by helping them complete 25% to 40% of code

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About Sela

Sela helps businesses achieve their digital transformation by implementing cloud solutions.

About Sela

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- Google Cloud (https://cloud.google.com/)
- Cloud GPUs (https://cloud.google.com/gpu/)
- Compute Engine (https://cloud.google.com/compute/)
- Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)
- Cloud IAM (https://cloud.google.com/iam/)
- Vertex AI (https://cloud.google.com/vertex-ai/)

Writing code is an essential part of being a developer. With recent breakthroughs in ML and the quantity and quality of open-source code availability, it has become possible to auto-complete much of the code developers need to spend time on writing. And that’s where the founders of Tabnine, Eran Yahav and Dror Weiss, saw an opportunity to help everyone involved in writing code. They created an AI-enabled tool for programmers to maintain a high level of productivity right from the start. Applying those models on private codebases of companies is something that Tabnine makes available, all while keeping the code safe and secure.

Tabnine has developed ML models that offer developers code suggestions, allowing them to autocomplete approximately 30 percent of their code. Powered by open-source and customized AI models, it helps to bridge any knowledge gaps. As a result, it helps companies and developers gain time and streamlines programmers' daily workflows.

"We envisioned that all software will be written with the use of AI in the near future, whether it was in creating software, in reviewing it or deploying it, and probably all three. When we started this sounded like a dream but now it is an everyday reality," says Eran Yahav, CTO & Co-Founder, Tabnine.

Delivering services for 1 million users with Google Cloud

With 1 million users in near real time, Tabnine has specific requirements to be able to run and deliver high-quality services. It cannot waste a second, literally. That's why its creators decided to use Cloud GPUs (https://cloud.google.com/gpu) (Graphics Processing Units). Their power allows for Tabnine to run important machine learning workloads and to achieve the low-latency needed to predict code almost instantaneously. Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) is the platform that incorporates these GPUs for faster scaling and serving.

"The availability and the price model for GPU instances offered by Google Cloud (https://cloud.google.com/) was more attractive than other providers," says Yahav. Tabnine also needed a cloud solution that supported the fast growth of their platform. Google Cloud offered a reliable infrastructure that allowed quick and easy auto-scaling for Tabnine's huge number of users. One of Tabnine's long-term aims is to make developers more productive. "We want to see the world develop twice as fast, twice as secure, and try to make everything two times better," says Brandon Jung, VP Ecosystem and Business Development, Tabnine.

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Improving consistency and security with Google Kubernetes Engine

One of the most valuable aspects of Tabnine is how it protects its users' privacy. It does so by only using license-free code to train its models and also by running on secure containers in Google Kubernetes Engine. Containers on Google Kubernetes Engine are isolated, which guarantees privacy and security. This means that users' custom AI models are never used to improve another client's code. At the same time, every Tabnine client benefits from open-source code to improve their programming, as well as their own private model. "It's critical to our customers that we can say 'your code is never shared with anyone'," says Jung.

Kubernetes presents another advantage for Tabnine when it comes to scaling business. Being open-source, Google Kubernetes Engine is a popular choice for customers to develop their own projects on. This increases the chances of Tabnine being consistent with its customers' creations and enables the code-autocompletion tool to run almost anywhere. "The Kubernetes cluster allows us to be dramatically faster and to implement our solutions more easily," explains Jung.

Tabnine also prides itself in being loyal to developers and friendly to the open-source community. "We appreciate that aspect of Kubernetes and how Google Cloud has a real commitment to open source," says Yahav.

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>
> *—Eran Yahav, CTO & Co-Founder, Tabnine*

Providing the right connections

Developing Tabnine required a lot of testing, and choosing the right products to build it was a delicate maneuver. "The Google Cloud team helped us connect with relevant product specialists, which helped us get the job done faster," explains Nir Marcu, VP R&D, Tabnine. Being connected to the right people helped Tabnine's team understand quickly what products would be relevant and how exactly Google Cloud could help them achieve their goals.

The ongoing collaboration with Google Cloud specialists helps Tabnine workers on a daily basis with anything from bug reporting to production issues, discussing business scaling and pricing. This way, Tabnine is able to find solutions to meet its needs faster. "Google Cloud gets the job done and is hassle free," says Yahav.

"Our number one mission at Tabnine is to serve developers and the community. We really want to be an integral part of it. We are developers. That's why we want to make sure that the next generation of developers gets the best we can offer them."

—Eran Yahav, Co-Founder, Tabnine

Helping the next generation of developers

The Tabnine team thinks that the future for developers lies in working even more closely with Google Cloud. "Our number one mission at Tabnine is to serve developers and the community," says Yahav. "We're developers, that's why we want to make sure that the next generation gets the best we can offer them. And Google Cloud will be an integral part of our work in the future."